

Notes: Bennett, Stulz, & Wang (JFE, 2020)

Does the Stock Market Make Firms More Productive?

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1 Big picture

- **Question.** Does better stock-market price discovery make firms more productive? The paper asks whether a more informative stock price raises firm-level total factor productivity rather than merely changing financing or valuation.
- **Core idea.** If stock prices embed more firm-specific information, managers, boards, activists, and investors can learn more from market signals. Those signals can then affect real decisions such as investment, CEO retention, and operating discipline.
- **Setting.** U.S. public firms, mainly 1994–2015, using Compustat, CRSP, TAQ, mutual fund holdings, ExecuComp, RiskMetrics, and Hoberg–Phillips competition data.
- **Main explanatory variable.** Stock price informativeness (SPI), measured primarily by the probability of informed trading (PIN) and stock price nonsynchronicity (PSI).
- **Main outcome.** Firm-level TFP, estimated from a production function using Compustat data and the Akerberg, Caves, and Frazer (2015) method.
- **Baseline result.** Firms with more informative stock prices are more productive. In the baseline fixed-effects regressions, a one-standard-deviation increase in PIN or PSI is associated with about a 5% increase in TFP in standard-deviation units.
- **Why this is not just an investment story.** The paper shows that the usual investment channel explains only a small portion of the SPI–TFP relation. The stock market appears to improve operating efficiency through broader managerial and governance channels.
- **Mechanism evidence.** When exogenous shocks make prices less informative, CEO turnover becomes less sensitive to Tobin’s q . If boards learn less from stock prices, they replace poorly performing CEOs less aggressively, which in turn lowers productivity.
- **Additional operating evidence.** More informative stock prices are associated with higher revenues, lower selling/general/administrative costs, and lower labor expenses.
- **Heterogeneity.** The SPI effect on productivity is stronger for smaller, younger, more focused, riskier, financially constrained, better-governed, and more competitive firms.
- **Why the paper matters.** It gives causal evidence that the stock market affects the real economy through information production, not just through capital raising. That is an important answer to the “stock market as sideshow” debate.

2 Data and measurement (institutional context and key objects)

2.1 Data sources and sample

- **Accounting data.** Compustat provides firm-level accounting variables used for productivity estimation and controls.
- **Stock-return data.** CRSP daily returns are used to construct PSI.
- **Trade and quote data.** TAQ is used to estimate PIN by classifying buys and sells with the Lee–Ready algorithm.

- **Mutual funds and institutions.** Thomson–Reuters mutual fund holdings and CRSP mutual fund data are used to build mutual-fund flow pressure; Thomson 13F provides institutional ownership and blockholders.
- **Management and governance.** ExecuComp provides CEO-turnover data; RiskMetrics provides governance variables such as the G-index.
- **Product-market competition.** Hoberg–Phillips text-based network industry measures provide HHI-style concentration, similarity, and fluidity.
- **Sample period.** The main sample runs from 1994 to 2015 because PIN requires TAQ data starting in 1993 and the regressions use lagged three-year averages of SPI.
- **Sample size.** The full main sample contains 66,341 firm-year observations.

2.2 Main productivity measure: firm-level TFP

- **Concept.** TFP measures the portion of output not explained by labor and capital inputs. Holding inputs fixed, higher TFP means the firm produces more.
- **Implementation.** The paper estimates a production function using the Akerberg, Caves, and Frazer (2015) method to address endogeneity in input choices.
- **Inputs and output.** Output is value added; capital is gross PP&E deflated by the private fixed-investment deflator and adjusted for asset age; labor is employment.
- **Alternative efficiency measures.** The paper also checks another firm-level TFP growth measure and several corporate-finance efficiency ratios, and the core result survives.

2.3 Main stock-price informativeness measures

- **PIN.** The probability of informed trading, estimated from a structural market microstructure model. High PIN means a larger fraction of order flow is informed.
- **PSI.** Stock price nonsynchronicity, derived from the fraction of firm returns not explained by market and industry returns. High PSI means more firm-specific return variation.
- **Additional measures.** The paper also uses Gamma (with and without market controls) and adjusted PIN (APIN) as robustness measures.
- **Interpretation.** In all of these measures, a higher number is intended to capture a more informative stock price.

2.4 Control variables and summary statistics

- **Baseline controls.** The main regressions control for log total assets, Tobin’s q , cash/assets, debt/assets, and R&D/assets.
- **Summary-statistic benchmarks.** Mean TFP is 0.03, mean PIN is 0.22, mean PSI is 2.22, mean log assets is 6.55, and mean Tobin’s q is 1.82.
- **Interpretation.** The firms in the sample are typical public corporations with substantial heterogeneity in size, leverage, growth opportunities, and information environment.

2.5 Event-study and mechanism datasets

- **Brokerage closures.** The authors hand-collect 17 brokerage-research-department closures from Kelly and Ljungqvist (2012), Bloomberg, Factiva, and IBES timing information.
- **S&P 500 additions.** Index additions are used as quasi-exogenous shocks to comovement and hence to firm-specific informativeness.
- **Mutual-fund flow pressure.** Following Edmans et al. (2012), the paper constructs hypothetical sales generated by mutual fund outflows to capture liquidity-driven trading pressure unrelated to fundamentals.
- **CEO turnover.** ExecuComp supplies CEO turnover events, allowing the paper to test whether price informativeness changes how boards react to market valuation.

3 Models and empirical specifications (all equations + interpretation)

3.1 Probability of informed trading (PIN)

On a trading day, informed traders arrive with probability-driven intensity $\alpha\mu$, while uninformed buy and sell orders arrive at rates ε_b and ε_s . The implied probability of information-based trading is

$$\text{PIN} = \frac{\alpha\mu}{\alpha\mu + (\varepsilon_b + \varepsilon_s)}. \quad (1)$$

- **Interpretation.** The numerator is the arrival rate of informed orders and the denominator is the arrival rate of all orders.
- **Economic intuition.** Higher PIN means a larger share of observed trading reflects private information, so stock prices should reveal more firm-specific information.

The daily likelihood of observing B_i buys and S_i sells is

$$\begin{aligned} L(\alpha, \delta, \mu, \varepsilon_b, \varepsilon_s \mid B_i, S_i) = & (1 - \alpha)P(B_i; \varepsilon_b)P(S_i; \varepsilon_s) \\ & + \alpha\delta P(B_i; \varepsilon_b)P(S_i; \mu + \varepsilon_s) \\ & + \alpha(1 - \delta)P(B_i; \mu + \varepsilon_b)P(S_i; \varepsilon_s), \end{aligned} \quad (2)$$

and the annual likelihood is

$$V = \prod_{i=1}^I L(\alpha, \delta, \mu, \varepsilon_b, \varepsilon_s \mid B_i, S_i). \quad (3)$$

- **Implementation.** The authors estimate the structural parameters by maximum likelihood using TAQ data aggregated to daily buy and sell counts.
- **Why it matters.** PIN has a micro-founded interpretation rather than being a pure reduced-form return-based measure.

3.2 Stock price nonsynchronicity (PSI)

For firm j in industry i , daily returns are decomposed as

$$r_{j,i,t} = \beta_{j,0} + \beta_{j,m}r_{m,t} + \beta_{j,i}r_{i,t} + \varepsilon_{j,i,t}, \quad (4)$$

where $r_{m,t}$ is the value-weighted market return and $r_{i,t}$ is the value-weighted industry return. The transformed nonsynchronicity measure is

$$\text{PSI}_j = \ln\left(\frac{1 - R_j^2}{R_j^2}\right). \quad (5)$$

- **Interpretation.** When market and industry returns explain less of a firm's return, R^2 is lower and PSI is higher.
- **Economic intuition.** A stock with more firm-specific information should move less one-for-one with broad market and industry factors.

3.3 Production function and the construction of TFP

The paper starts from a Cobb–Douglas production function,

$$Y = AK^\alpha L^\beta, \quad (6)$$

which in logs becomes

$$\ln(Y) = \alpha \ln(K) + \beta \ln(L) + \ln(A). \quad (7)$$

The estimating equation is

$$\ln(Y) = \beta_0 + \alpha \ln(K) + \beta \ln(L) + \varepsilon, \quad (8)$$

so measured productivity is

$$\ln(A) = \beta_0 + \varepsilon. \quad (9)$$

In panel notation,

$$y_{i,t} = \beta_0 + \beta_k k_{i,t} + \beta_l l_{i,t} + \omega_{i,t}, \quad (10)$$

and the fitted TFP measure is

$$\text{TFP}_{i,t} = y_{i,t} - \hat{\beta}_k k_{i,t} - \hat{\beta}_l l_{i,t}. \quad (11)$$

- **Interpretation.** The residual from the production function is the firm's measured efficiency.
- **Why ACF matters.** Input choices are endogenous, so OLS production-function estimates are biased; the Akerberg–Caves–Frazer procedure is designed to address this problem.

3.4 Baseline panel regression of productivity on stock price informativeness

The core estimating equation is

$$TFP_{i,t} = \beta_0 + \beta_1 SPI_{i,t-3,t-1} + X_{i,t}\Gamma + \mu_i + \vartheta_t + \varepsilon_{i,t}, \quad (12)$$

where $SPI_{i,t-3,t-1}$ is the average of the stock-price informativeness measure over the previous three years.

- **Regressor of interest.** SPI is either PIN, PSI, Gamma, or APIN.
- **Fixed effects.** Firm fixed effects absorb time-invariant firm characteristics; year fixed effects absorb aggregate shocks.
- **Expected sign.** $\beta_1 > 0$: more informative stock prices should improve managerial learning and raise productivity.
- **Why lag SPI?** Using a moving average of lagged informativeness reduces simultaneity and reverse-causality concerns.

3.5 Brokerage-house closures as exogenous shocks to informativeness

For the difference-in-differences analysis around brokerage closures, the main post-treatment regression is

$$TFP_{i,t} = \beta_0 + \beta_1 Treatment_post_{i,t} + X_{i,t}\Gamma + \mu_i + \vartheta_t + \varepsilon_{i,t}, \quad (13)$$

where $Treatment_post$ equals one for treated firms in the first four years after a brokerage closure and zero otherwise.

The dynamic version is

$$TFP_{i,t} = \beta_0 + \sum_{k=1}^4 \beta_k I_{k,i,t} + X_{i,t}\Gamma + \mu_i + \vartheta_t + \varepsilon_{i,t}, \quad (14)$$

where $I_{k,i,t}$ indicates year k after a brokerage closure.

- **Interpretation.** Brokerage closures reduce analyst-produced information about treated firms, so they should reduce SPI.
- **Identification logic.** Research-department shutdowns are plausibly unrelated to the fundamentals of the covered firms, making them a useful source of exogenous variation.

3.6 S&P 500 additions as exogenous shocks to informativeness

For S&P 500 additions, the post-treatment specification is

$$TFP_{i,t} = \beta_0 + \beta_1 SP500_addition_{i,t} + X_{i,t}\Gamma + \mu_i + \vartheta_t + \varepsilon_{i,t}, \quad (15)$$

and the dynamic event-time version is

$$\text{TFP}_{i,t} = \gamma_0 + \sum_{k=1}^4 \gamma_k I_{k,i,t} + X_{i,t} \Gamma + \mu_i + \vartheta_t + \varepsilon_{i,t}. \quad (16)$$

- **Interpretation.** Addition to the index increases comovement with the index and reduces the firm-specific information content of the stock price.
- **Expected sign.** If lower SPI reduces productivity, the treatment coefficients should be negative.

3.7 Mutual-fund redemption pressure

To show that liquidity-driven selling pressure makes prices less informative, the paper estimates

$$\text{SPI}_{i,t} = \beta_0 + \beta_1 \text{MFFlow}_{i,t-1} + X_{i,t-1} \Gamma + \mu_i + \vartheta_t + \varepsilon_{i,t}, \quad (17)$$

where MFFlow equals one if hypothetical mutual-fund sales for the stock are positive.

The direct reduced-form effect on productivity is estimated via

$$\text{TFP}_{i,t} = \beta_0 + \beta_1 \text{MFFlow}_{i,t-1} + X_{i,t} \Gamma + \mu_i + \vartheta_t + \varepsilon_{i,t}. \quad (18)$$

- **Interpretation.** Redemptions create price pressure unrelated to fundamentals, increasing noise trading and discouraging information acquisition.
- **Expected sign.** The first-stage effect on SPI should be negative, and the reduced-form effect on TFP should also be negative.

3.8 CEO-turnover mechanism: mutual-fund flow pressure

The CEO-turnover regression is

$$\begin{aligned} \text{Turnover}_{i,t} = & \beta_0 + \beta_1 (q_{i,t-1} \times \text{MFFlow}_{i,t-2}) + \beta_2 q_{i,t-1} + \beta_3 \text{MFFlow}_{i,t-2} \\ & + X_{i,t-1} \Gamma + \mu_j + \vartheta_t + \varepsilon_{i,t}. \end{aligned} \quad (19)$$

- **Dependent variable.** A dummy equal to one if the firm experiences CEO turnover that year.
- **Key coefficients.** $\beta_2 < 0$ means low market valuation increases turnover; $\beta_1 > 0$ means flow pressure weakens that sensitivity.
- **Economic interpretation.** If prices become noisier, boards should rationally put less weight on them when judging CEOs.

3.9 CEO-turnover mechanism: brokerage closures

Using brokerage closures instead of mutual-fund pressure, the paper estimates

$$\begin{aligned} \text{Turnover}_{i,t} = & \beta_0 + \beta_1 (q_{i,t-1} \times \text{Closure_post}_{i,t}) + \beta_2 q_{i,t-1} + \beta_3 \text{Closure_post}_{i,t} \\ & + X_{i,t-1} \Gamma + \mu_j + \vartheta_t + \varepsilon_{i,t}. \end{aligned} \quad (20)$$

- **Interpretation.** This is the same logic with a different shock to informativeness.
- **Expected sign.** Again, the interaction term should offset the baseline negative q -turnover relation.

3.10 Does turnover improve productivity?

To complete the mechanism, the paper estimates whether CEO replacement precedes productivity improvements:

$$\Delta TFP_{i,t} = \beta_0 + \beta_1 Turnover_{i,t-1} + \beta_2 Turnover_{i,t-2} + X_{i,t}\Gamma + \mu_i + \vartheta_t + \varepsilon_{i,t}. \quad (21)$$

- **Interpretation.** If replacing a CEO raises later productivity, then lower turnover caused by noisier prices is a plausible channel from SPI to TFP.
- **Timing intuition.** Productivity should react with a lag because new CEOs need time to change operations.

3.11 Inputs, outputs, and operating channels

The paper studies real operating margins with

$$IO_{i,t} = \beta_0 + \beta_1 SPI_{i,t-3,t-1} + X_{i,t}\Gamma + \mu_i + \vartheta_t + \varepsilon_{i,t}, \quad (22)$$

where IO is alternatively log revenue, $SG\&A/Assets$, or log labor cost.

- **Expected sign.** More informative prices should raise output and reduce wasteful input use.
- **Economic meaning.** This turns the abstract SPI–TFP relation into concrete operational margins: higher sales, lower overhead, and lower labor expense for comparable firms.

3.12 Cross-sectional heterogeneity by firm characteristics

For any firm characteristic $F_{i,t}$, the generic interaction regression is

$$TFP_{i,t} = \beta_0 + \beta_1 (F_{i,t} \times SPI_{i,t-3,t-1}) + \beta_2 SPI_{i,t-3,t-1} + \beta_3 F_{i,t} + X_{i,t}\Gamma + \mu_i + \vartheta_t + \varepsilon_{i,t}. \quad (23)$$

The paper uses this form for size, age, diversification, and business risk.

- **Predictions.** The interaction should be negative for large, old, and diversified firms, and positive for risky firms.
- **Intuition.** Some firms are less able to extract or act on information in their prices because of bureaucracy or complexity.

3.13 Cross-sectional heterogeneity by financial constraints

For financial constraints, the paper estimates

$$\begin{aligned} \text{TFP}_{i,t} = & \beta_0 + \beta_1 (FC_{i,t} \times SPI_{i,t-3,t-1}) + \beta_2 SPI_{i,t-3,t-1} \\ & + \beta_3 FC_{i,t} + X_{i,t}\Gamma + \mu_i + \vartheta_t + \varepsilon_{i,t}. \end{aligned} \quad (24)$$

- **Measures.** No-dividend, Whited–Wu, no bond rating, and Kaplan–Zingales indexes.
- **Expected sign.** Usually positive: constrained firms have a lot to gain from better information if it helps them allocate scarce resources.

3.14 Cross-sectional heterogeneity by product-market competition

For competition, the paper estimates

$$\begin{aligned} \text{TFP}_{i,t} = & \beta_0 + \beta_1 (\text{Competition}_{i,t} \times SPI_{i,t-3,t-1}) + \beta_2 SPI_{i,t-3,t-1} \\ & + \beta_3 \text{Competition}_{i,t} + X_{i,t}\Gamma + \mu_i + \vartheta_t + \varepsilon_{i,t}. \end{aligned} \quad (25)$$

- **Measures.** Low HHI, high product similarity, and high fluidity from Hoberg–Phillips text data.
- **Expected sign.** Positive: firms facing stronger competition have stronger incentives to use market information efficiently.

3.15 Cross-sectional heterogeneity by governance

For governance, the paper estimates

$$\begin{aligned} \text{TFP}_{i,t} = & \beta_0 + \beta_1 (Gov_{i,t} \times SPI_{i,t-3,t-1}) + \beta_2 SPI_{i,t-3,t-1} \\ & + \beta_3 Gov_{i,t} + X_{i,t}\Gamma + \mu_i + \vartheta_t + \varepsilon_{i,t}. \end{aligned} \quad (26)$$

- **Measures.** High institutional ownership, the log number of blockholders, and the G-index.
- **Expected sign.** Positive for stronger monitoring variables and negative for the G-index, which proxies for weak governance.

3.16 Alternative productivity-growth measure

The appendix also studies a TFP-growth measure defined as

$$\Delta \ln(\text{TFP}_{i,t}) = \Delta \ln(Y_{i,t}) - \frac{1}{2} [S_{L,i,t} + S_{L,i,t-1}] \Delta \ln(L_{i,t}) - \frac{1}{2} [S_{K,i,t} + S_{K,i,t-1}] \Delta \ln(K_{i,t}), \quad (27)$$

where S_L and S_K are labor and capital cost shares.

- **Purpose.** This is an alternative productivity metric from Chun et al. (2011, 2016).
- **Why useful.** It checks that the main result is not specific to one production-function estimator.

4 Identification and instruments (what makes the estimates credible)

4.1 Baseline identification idea

- The baseline specification compares the same firm to itself over time, holding fixed all time-invariant firm characteristics through firm fixed effects.
- The use of a three-year lagged average of SPI reduces concerns that current productivity mechanically drives current stock-price informativeness.
- Year fixed effects remove macroeconomic shocks that affect both the stock market and firm performance.
- The main remaining concern is time-varying omitted variables, such as technology shocks, that could simultaneously raise productivity and attract informed trading.

4.2 Brokerage-house closures

- **Relevance.** Closure of a brokerage research department reduces analyst coverage and information production for covered stocks.
- **Evidence.** The closure dummy reduces PIN by 0.013 and PSI by 0.101, both statistically significant.
- **Exclusion logic.** The shutdown of a broker's research division is plausibly driven by the broker's business conditions rather than by the fundamentals of each covered firm.
- **Design.** The paper uses a propensity-score-matched sample and a $[-4, +4]$ event window around closures.

4.3 S&P 500 additions

- **Relevance.** Once a stock enters the S&P 500, it comoves more with the index, so less of its return variation is firm-specific.
- **Evidence.** The addition dummy lowers PIN by about 0.008 and PSI by about 0.154.
- **Exclusion logic.** Index inclusion is committee-based and largely outside the direct control of the treated firms, so it is commonly used as a quasi-natural experiment.
- **Caveat handled in the paper.** The authors show the result remains after matching on pre-trends, stock returns, or lagged TFP, and after controlling for liquidity and institutional ownership.

4.4 Mutual-fund redemption pressure

- **Relevance.** Fund outflows generate liquidity-driven hypothetical sales that push prices for non-fundamental reasons.

- **Evidence.** MFFlow lowers PIN by 0.005, lowers PSI by 0.137, and reduces TFP by about 0.012 to 0.013.
- **Exclusion logic.** Because the measure is based on hypothetical sales implied by fund outflows, it is less contaminated by active fund-manager information than actual sales would be.
- **Interpretation.** If non-fundamental trading pressure lowers productivity, that supports a causal chain running from informativeness to real outcomes.

4.5 Mechanism identification through CEO turnover

- The mechanism tests are stronger than a simple reduced-form result because they show *how* a decline in informativeness changes managerial discipline.
- The board should rationally use stock prices less when those prices contain less firm-specific information.
- The paper then closes the mechanism by showing that CEO turnover two years earlier predicts a significant increase in ΔTFP .

4.6 Inference and remaining limitations

- Robust standard errors are clustered at the firm level in the main panel regressions.
- The paper does not claim that CEO turnover is the only channel; it presents it as one concrete and empirically verifiable mechanism.
- A remaining limitation is that SPI may still proxy for some broader information environment, but the multiple quasi-experimental shocks make a purely omitted-variable explanation much harder to maintain.

5 Tables and figures

5.1 Table 1: Summary statistics

Read:

- The sample is large and heterogeneous: 66,341 firm-years with wide dispersion in TFP, leverage, cash, and Tobin's q .
- Mean PIN of 0.22 and mean PSI of 2.22 are in line with prior literature, so the informativeness measures look well behaved.
- The large dispersion in TFP and SPI is what makes the paper's cross-sectional tests meaningful.

5.2 Table 2: Price informativeness and productivity

Read:

Table 1

Summary statistics.

This table presents summary statistics for TFP, stock price informativeness measures PIN and PSI, and firm characteristics. The sample consists of firms in Compustat for which TFP and the stock price informativeness measures are available for the years 1994–2015, inclusive. All variables are winsorized at the 1st and 99th percentile values. Variable definitions are in [Appendix A](#).

Variable	Mean	p25	p50	p75	SD	N
TFP	0.03	-0.33	0.00	0.37	0.60	66,341
PIN	0.22	0.14	0.20	0.28	0.11	66,341
PSI	2.22	0.90	2.06	3.44	1.71	63,504
Log (assets)	6.55	5.08	6.43	7.88	2.00	66,341
Cash/assets	0.14	0.02	0.08	0.21	0.17	66,341
Debt/assets	0.24	0.05	0.21	0.36	0.22	66,134
R&D/assets	0.03	0.00	0.00	0.03	0.06	66,341
Tobin's q	1.82	1.10	1.41	2.03	1.40	64,876
PP&E/assets	0.28	0.09	0.21	0.42	0.23	66,341
Business risk	0.03	0.02	0.03	0.04	0.02	55,492
Log (N_blockholders)	1.06	0.68	1.10	1.39	0.55	25,511
Diversified	0.42	0	0	1	0.49	66,341
SG&A/assets	0.25	0.07	0.19	0.35	0.25	66,341
G-index	8.94	7	9	11	2.74	19,796

Table 2

Price informativeness and productivity.

This table presents panel regressions of total factor productivity (TFP) on stock price informativeness and other firm-level controls. In Panel A, stock price informativeness is measured by the probability of informed trading (PIN) and stock price nonsynchronicity (PSI). IQS is investment-q sensitivity. In Panel B, we test additional SPI measures. The first measure is Gamma, a trading-based informativeness measure calculated in Eq. (12) in [Llorente et al. \(2002\)](#). We calculate this measure in two ways. The first method (Columns 1 and 4) is as in Eq. (3) in [Fildes \(2002\)](#) and controls for both firm and market returns, while the second method (Columns 2 and 5) only controls for firm returns as in the original [Llorente et al. \(2002\)](#). The last additional stock price informativeness measure, Adjusted PIN (APIN), is calculated using Eq. (7) in [Duarte and Young \(2009\)](#). In our regressions, we use the average SPI over the previous three years. All specifications include firm and year fixed effects. The sample consists of firms in Compustat for which TFP and the stock price informativeness measures are available for the years 1994–2015 except for Column 5 in Panel A, which is from 1962 to 2015. Robust standard errors are clustered at the firm level. Variable definitions are in [Appendix A](#). ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively.

Panel A: Primary SPI measures (PIN & PSI)						
Variables	(1) TFP	(2) TFP	(3) TFP	(4) TFP	(5) TFP	(6) TFP
PIN	0.262*** [8.80]		0.256*** [8.63]			0.234*** [7.82]
PSI		0.018*** [5.92]		0.019*** [6.31]	0.010*** [3.82]	0.020*** [6.50]
Log (assets)	0.235*** [35.75]	0.238*** [34.41]	0.225*** [33.39]	0.228*** [32.15]	0.170*** [30.02]	0.225*** [31.49]
Tobin's q	0.079*** [19.63]	0.079*** [19.34]	0.077*** [19.30]	0.078*** [19.03]	0.136*** [43.68]	0.077*** [17.11]
Cash/assets			0.042 [1.54]	0.043 [1.52]	-0.049* [-1.85]	0.067** [2.33]
Debt/assets			-0.226*** [-10.80]	-0.235*** [-10.88]	-0.278*** [-14.83]	-0.274*** [-9.91]
R&D/assets			-1.139*** [-9.76]	-1.137*** [-9.72]	-1.943*** [-15.84]	-1.153*** [-9.13]
Capex/assets						0.658*** [9.50]
IQS						-0.008* [-1.90]
Capex/assets* IQS						0.060 [1.43]
Observations	61,554	58,889	61,363	58,700	108,832	60,121
R-squared	0.178	0.176	0.192	0.191	0.172	0.172
Firm FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y
Panel B: Additional SPI measures (Gamma and APIN)						
Variables	(1) TFP	(2) TFP	(3) TFP	(4) TFP	(5) TFP	(6) TFP
Gamma (market)	0.035*** [2.65]			0.031*** [2.35]		
Gamma (No market)		0.028** [2.32]			0.024** [2.05]	
APIN			0.177*** [5.55]			0.175*** [5.55]
Log (assets)	0.226*** [33.23]	0.226*** [33.22]	0.244*** [34.55]	0.217*** [31.50]	0.217*** [31.50]	0.234*** [32.21]
Tobin's q	0.078*** [18.32]	0.078*** [18.32]	0.074*** [18.15]	0.077*** [18.08]	0.077*** [18.09]	0.072*** [17.84]
Cash/assets			0.026 [0.90]		0.026 [0.90]	0.039 [1.34]
Debt/assets			-0.256*** [-11.68]		-0.256*** [-11.69]	-0.259*** [-11.69]
R&D/assets			-1.161*** [-9.39]		-1.161*** [-9.39]	-1.171*** [-9.29]
Observations	54,485	54,485	48,924	54,310	54,310	48,760
R-squared	0.174	0.174	0.179	0.190	0.190	0.195
Firm FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y

- This is the core table. With firm and year fixed effects, both PIN and PSI enter positively and very significantly in baseline TFP regressions.
- The main coefficients are about 0.256 for PIN and 0.019 for PSI with the full control set.
- Adding investment variables barely changes the SPI coefficients, so the usual investment-efficiency channel explains only a small part of the productivity effect.
- Gamma and adjusted PIN are also positive, showing the result is not tied to one measure of informativeness.

5.3 Table 3: The effect of brokerage-house closures on stock price informativeness

Read:

- Brokerage closures significantly reduce both PIN and PSI, which establishes the first stage for the closure-based quasi-experiment.
- The treatment lowers PIN by 0.013 and PSI by 0.101.
- This supports the idea that analyst-information production is one source of informative prices.

5.4 Figure 1: Graphical analysis of brokerage closures and TFP

Read:

- The event-study graph shows no meaningful pre-trend differences between treated and matched control firms before closures.

Table 3

The effect of brokerage house closures on stock price informativeness. This table shows the effect of brokerage closures on stock price informativeness. The specification is as follows: $SPI_{it} = \beta_0 + \beta_1 \cdot Closure_{it} + X_{i,t-1} \cdot \Gamma + \mu_i + \vartheta_t + \varepsilon_{it}$. SPI is PIN or PSI. Closure is a dummy variable that equals one if a stock is covered by a closed research department in the previous one or two years and zero otherwise. Control variables are the same as used in Column 3 of Table 2. The sample consists of firms in Compustat for which the stock price informativeness measures are available for the years 1994–2015. Firm and year fixed effects are included. Robust standard errors are clustered at the firm level. ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively.

Variables	(1) PIN	(2) PSI
Closure	-0.013** [-2.48]	-0.101** [-1.97]
Log (assets)	-0.032*** [-22.06]	-0.469*** [-25.04]
Tobin's q	-0.005*** [-5.74]	-0.278*** [-34.37]
Cash/assets	-0.008 [-1.20]	-0.190** [-2.51]
Debt/assets	0.035*** [6.48]	0.784*** [10.90]
R&D/assets	-0.001 [-0.04]	0.426* [1.71]
Observations	44,359	42,257
R-squared	0.484	0.765
Firm FE	Y	Y
Year FE	Y	Y

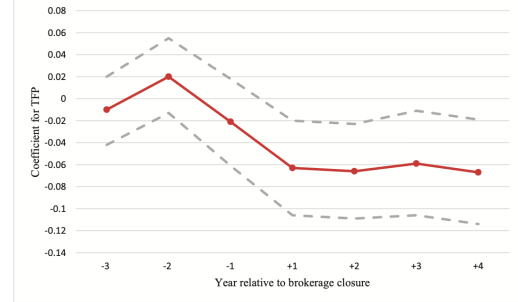


Fig. 3. Graphical analysis of the effect of brokerage closures on TFP. This figure shows the graphical analysis for the effect of brokerage closures on TFP. The y-axis plots the coefficient estimates from regressing TFP on dummy variables indicating the year relative to a brokerage closure, controlling for year fixed effects and firm size. The x-axis shows the time relative to the brokerage closure. The dashed lines correspond to 90% confidence intervals of the coefficient estimates, and the confidence intervals are based on standard errors clustered at the firm level. The sample includes firms that experienced brokerage closures and control firms matched by total assets and Tobin's q in the same industry (two-digit SIC code).

- After the closure, treated firms experience a persistent drop in TFP relative to controls.
- This visual evidence strengthens the causal interpretation of the DiD estimates in Table 4.

5.5 Table 4: DiD analysis around brokerage-house closures

Read:

- The post-treatment coefficient is about -0.043 , implying that TFP falls after research-department closures.
- Event-time effects are negative in years 1 through 4 after closure: roughly -0.051 , -0.049 , -0.040 , and -0.043 .
- The effect attenuates slightly over time, consistent with coverage and information production gradually recovering.

5.6 Table 5: The effect of S&P 500 additions on stock price informativeness

Read:

- S&P 500 additions reduce PIN and PSI, consistent with the idea that index inclusion raises comovement and lowers firm-specific price discovery.
- The first-stage effects are economically meaningful, especially for PSI.
- This table motivates the second quasi-experimental design.

Table 4

DiD analysis: brokerage house closures and productivity.

This table shows DiD tests based on the closures of brokerage house research departments. The sample is from 1996 to 2011. A firm is defined as a treated firm if its stock is covered by a closed research department. For each closure event, we define an event window as four years before to four years after the closure. For each treated firm, we use propensity score matching to choose a control firm in the same industry (two-digit SIC) and matched by total assets and Tobin's q using Mahalanobis distance. In Model 1 the treatment dummy *Treatment_post* equals one if a stock is covered by a closed research department and the year is between one and four years after the closure year and zero otherwise. In Model 2 we define four treatment dummy variables, one dummy for each year during the four years after a closure. Closure years are dropped in the regressions. All specifications include firm fixed effects and year fixed effects. Robust standard errors are clustered at the firm level. Variable definitions are in [Appendix A](#). ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively.

Variables	(1) TFP	(2) TFP
<i>Treatment_post</i>	-0.043** [-2.29]	
One year after closure		-0.051*** [-2.75]
Two years after closure		-0.049** [-2.45]
Three years after closure		-0.040* [-1.80]
Four years after closure		-0.043* [-1.94]
Log (assets)	0.248*** [12.88]	0.249*** [12.98]
Tobin's q	0.035*** [3.85]	0.035*** [3.85]
Cash/assets	0.173** [2.27]	0.173** [2.27]
Debt/assets	-0.248*** [-4.76]	-0.246*** [-4.71]
R&D/assets	-1.364*** [-3.23]	-1.361*** [-3.22]
Observations	7,851	7,851
R-squared	0.801	0.801
Firm FE	Y	Y
Year FE	Y	Y

Table 5

The effect of S&P 500 additions on stock price informativeness.

This table shows the effect of S&P 500 additions on the stock price informativeness. The specification is as follows: $SPI_{it} = \beta_0 + \beta_1 \cdot Addition_{it} + X_{i,t-1} \cdot \Gamma + \mu_i + \vartheta_t + \varepsilon_{it}$. *SPI* is *PIN* or *PSI*. *Addition* is a dummy variable that equals one if a firm is added to the S&P 500 index in the previous one or two years and zero otherwise. Control variables are the same as used in Column 3 of [Table 2](#). Firm and year fixed effects are included. The sample includes firms with above yearly median book assets because firms added into S&P 500 index are unlikely to have assets below median assets. Robust standard errors are clustered at the firm level. ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively.

Variables	(1) PIN	(2) PSI
<i>Addition</i>	-0.008* [-1.66]	-0.154*** [-3.31]
Log (assets)	-0.019*** [-9.80]	-0.297*** [-10.45]
Tobin's q	-0.002** [-2.17]	-0.106*** [-5.21]
Cash/assets	-0.011 [-1.28]	-0.275** [-2.53]
Debt/assets	0.041*** [4.91]	0.848*** [6.93]
R&D/assets	0.001 [0.02]	0.220 [0.47]
Observations	21,830	20,913
R-squared	0.356	0.726
Firm FE	Y	Y
Year FE	Y	Y

5.7 Figure 2: Graphical analysis of S&P 500 additions and TFP

Read:

- The graph again shows no obvious pre-trend differences before treatment.
- After index inclusion, treated firms experience lower TFP relative to matched controls.
- The pattern becomes more negative as event time advances, foreshadowing the dynamic estimates in Table 6.

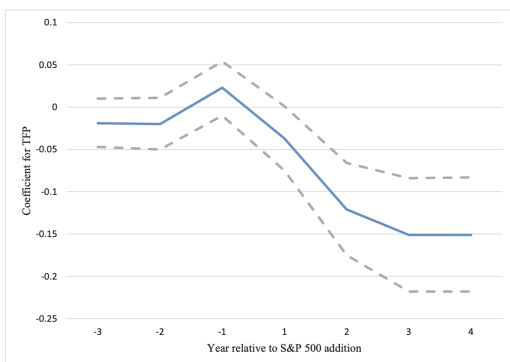


Fig. 2. Graphical analysis of the effect of S&P 500 index additions on TFP. This figure shows the graphical analysis for the effect of S&P 500 index additions on TFP. The y-axis plots the coefficient estimates from regressing TFP on dummy variables indicating the year relative to an index addition, controlling for year fixed effects and firm size. The x-axis shows the time relative to the index addition. The dashed lines correspond to 90% confidence intervals of the coefficient estimates, and the confidence intervals are based on standard errors clustered at the firm level. The sample includes firms that were added to the S&P 500 index and control firms matched by total assets and Tobin's q in the same industry (two-digit SIC code).

Table 6

DiD analysis: S&P 500 index additions and productivity.

This table shows DiD tests based on S&P 500 index additions. A firm is defined as a treated firm if it is added to the S&P 500 index in a year. For each index addition, we define an event window as four years before to four years after the index addition. For each treated firm, we use propensity score matching to choose a control firm in the same industry (two-digit SIC) and matched by total assets and Tobin's q with minimum Mahalanobis distance in Models 1, 4, 5, and 6. We use additional match variable lagged stock return in Model 2 or lagged TFP in Model 3. In Models 1 to 5 the treatment dummy, *SP500_addition*, equals one if a firm is added to the S&P 500 index over the previous four years and zero otherwise. In Model 6 we define four treatment dummy variables, one dummy for each year during the four years after a closure. All specifications include firm and year fixed effects. Robust standard errors are clustered at the firm level. Variable definitions are in Appendix A. ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively.

Variables	(1) TFP	(2) TFP	(3) TFP	(4) TFP	(5) TFP	(6) TFP
SP500_addition	-0.083*** [-3.81]	-0.100*** [-4.78]	-0.082*** [-4.00]	-0.084*** [-3.77]	-0.058** [-2.16]	
One year after add						-0.036* [-1.86]
Two years after add						-0.105*** [-3.87]
Three years after add						-0.141*** [-4.42]
Four years after add						-0.146*** [-4.44]
Log (assets)	0.167*** [7.26]	0.208*** [8.31]	0.195*** [9.64]	0.192*** [7.22]	0.197*** [5.94]	0.183*** [7.43]
Tobin's q	0.049*** [6.38]	0.052*** [5.11]	0.057*** [6.82]	0.043*** [6.01]	0.060*** [4.66]	0.086*** [6.18]
Cash/assets	-0.025 [-0.24]	0.080 [0.71]	0.138 [1.34]	-0.035 [-0.27]	-0.085 [-0.62]	-0.077 [-0.68]
Debt/assets	-0.288*** [-3.57]	-0.253*** [-2.97]	-0.302*** [-4.85]	-0.248*** [-3.15]	-0.232*** [-2.56]	-0.243*** [-2.83]
R&D/assets	-1.207*** [-2.28]	-1.086*** [-2.26]	-0.980*** [-1.92]	-0.690 [-1.30]	-0.433 [-1.00]	-1.047*** [-2.01]
Amihud				-0.098*** [-2.65]	-1.453*** [-6.64]	
Stock return				0.017 [1.03]	-0.009 [-0.59]	
Inst ownership					0.071 [0.73]	
Observations	3,908	3,855	3,887	3,202	2,141	3,482
R-squared	0.193	0.210	0.230	0.176	0.178	0.191
Firm FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y

5.8 Table 6: DiD analysis around S&P 500 additions

Read:

- The basic post-treatment effect is large and negative: around -0.083 in the baseline matched sample.
- The result remains when control firms are matched on lagged stock returns or lagged TFP, and when liquidity, returns, and institutional ownership are added as controls.
- Event-time effects deepen from about -0.036 in year 1 to around -0.146 by year 4, indicating a persistent productivity cost of lower informativeness.

5.9 Table 7: Mutual-fund flow pressure and stock price informativeness

Read:

- Liquidity-driven fund-flow pressure significantly lowers both PIN and PSI.
- This provides a third source of quasi-exogenous variation in informativeness.
- Because the shock is unrelated to firm fundamentals by construction, the table is a strong first-stage result.

Table 7

Mutual fund flow pressure and stock price informativeness.

This table presents the estimates of the specification $SPI_{it} = \beta_0 + \beta_1 \cdot MFFlow_{i,t-1} + X_{i,t-1} \cdot \Gamma + \mu_i + \vartheta_t + \varepsilon_{it}$. SPI is PIN or PSI. MFFlow is a dummy variable that equals one if a stock's hypothetical fund sales is positive and zero otherwise. The hypothetical fund sales are constructed as in [Edmans et al. \(2012\)](#). The sample consists of firms in Compustat for which the stock price informativeness measures are available for the years 1994–2015. Firm fixed effects and year fixed effects are included. Robust standard errors are clustered at the firm level. Variable definitions are in [Appendix A](#). ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively.

Variables	(1) PIN	(2) PSI
MFFlow	-0.005*** [-3.75]	-0.137*** [-8.67]
Log (assets)	-0.032*** [-21.92]	-0.439*** [-22.82]
Tobin's q	-0.005*** [-5.72]	-0.136*** [-7.13]
Cash/assets	-0.007 [-1.16]	-0.341*** [-4.25]
Debt/assets	0.034*** [6.37]	0.840*** [10.93]
R&D/assets	-0.001 [-0.05]	0.400 [1.53]
Observations	44,359	42,257
R-squared	0.484	0.760
Firm FE	Y	Y
Year FE	Y	Y

Table 8

Mutual fund redemption pressure and TFP.

This table shows the effect of mutual fund redemption on TFP. The specification is $TFP_{it} = \beta_0 + \beta_1 \cdot MFFlow_{i,t-1} + X_{it} \cdot \Gamma + \mu_i + \vartheta_t + \varepsilon_{it}$. MF-Flow is a dummy variable that equals one if a stock's hypothetical fund sales is positive and zero otherwise. The hypothetical fund sales follow that in [Edmans et al. \(2012\)](#). The sample consists of firms in Compustat for which our TFP variable is available for the years 1994–2015. Firm fixed effects and year fixed effects are controlled. Robust standard errors are clustered at the firm level. Variable definitions are in [Appendix A](#). ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively.

Variables	(1) TFP	(2) TFP
MFFlow	-0.012*** [-2.93]	-0.013*** [-3.15]
Log (assets)	0.217*** [36.11]	0.209*** [33.92]
Tobin's q	0.128*** [42.33]	0.126*** [42.00]
Debt/assets		-0.207*** [-10.95]
Cash/assets		0.018 [0.69]
R&D/assets		-1.196*** [-10.85]
Observations	67,572	67,364
R-squared	0.791	0.794
Firm FE	Y	Y
Year FE	Y	Y

5.10 Table 8: Mutual-fund redemption pressure and TFP

Read:

- The mutual-fund flow indicator enters negatively and significantly in TFP regressions, with coefficients near -0.012 to -0.013 .
- This is a direct reduced-form link from a non-fundamental decline in informativeness to lower firm productivity.

5.11 Table 9: Exogenous shocks to informativeness and CEO turnover

Read:

- In Panel A, the interaction $MFFlow \times q$ is positive while the coefficient on q is negative, meaning CEO turnover becomes less sensitive to firm value when prices are noisier.
- In Panel B, the same pattern appears around brokerage closures: the turnover- q sensitivity is weaker after closures.
- This is the paper's cleanest mechanism result: boards learn less from stock prices when stock prices contain less information.

5.12 Table 10: CEO turnover and later improvements in TFP

Read:

- CEO turnover two years earlier predicts a significant increase in ΔTFP of about 0.012 to 0.013.

Table 9

Exogenous shocks to price informativeness and CEO turnover.

This table illustrates how decreases in stock price informativeness affect CEO turnover sensitivity to q . Panel A (B) presents the estimates of the logit model, where mutual fund flow pressure (brokerage house closure) is the exogenous shock to price informativeness. Turnover is a dummy variable that equals one if a firm experiences a CEO turnover in the year and zero otherwise. In Panel A, MFFlow is a dummy variable that equals one if a stock's hypothetical fund sales is positive and zero otherwise. The hypothetical fund sales are constructed as in [Edmans et al. \(2012\)](#). The sample consists of firms in Execucomp for the years 1994–2015. In Panel B, Closure_post is a dummy variable that equals one if a firm experienced a brokerage closure in the previous four years and zero otherwise. In Panel B, we use the window $[-4, 4]$ around brokerage closures and use a PSM sample as in [Table 4](#) for which Execucomp data is available. Old CEO is a dummy variable that equals one if a CEO's age is above 60 and zero otherwise. Industry fixed effects and year fixed effects are included, where industry classification is the one-digit SIC. Robust standard errors are clustered at the firm level. Variable definitions are in [Appendix A](#). ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively.

Panel A: CEO turnover sensitivity to q and mutual fund flow pressure		
Variables	(1) CEO turnover	(2) CEO turnover
MFFlow $\times$ tobin's q	0.130*** [2.80]	0.126*** [2.84]
Tobin's q	-0.211*** [-5.40]	-0.078** [-2.01]
MFFlow	-0.353*** [-3.68]	-0.347*** [-3.74]
Log (assets)	0.086*** [5.02]	0.082*** [4.69]
Return volatility	8.545*** [4.21]	3.253 [1.43]
ROA		-3.787*** [-13.77]
Debt/Assets		-0.368** [-2.46]
Old CEO		0.862*** [19.20]
Observations	21,148	20,691
R-squared	0.0110	0.0461
Industry FE	Y	Y
Year FE	Y	Y
Panel B: CEO turnover sensitivity to q and brokerage research department closures		
Variables	(1) CEO turnover	(2) CEO turnover
Tobin's $q \times$ closure_post	0.255* [1.86]	0.270** [1.98]
Tobin's q	-0.236*** [-3.57]	-0.168** [-2.25]
Closure_post	-0.423 [-1.34]	-0.432 [-1.36]
Log (assets)	0.044 [1.00]	0.062 [1.39]
Volatility	5.009 [1.01]	-0.005 [-0.00]
ROA		-2.068*** [-3.03]
Debt/assets		-0.270 [-0.73]
Old CEO		0.006 [0.05]
Observations	3,410	3,371
R-squared	0.0132	0.0177
Industry FE	Y	Y
Year FE	Y	Y

Table 10

CEO turnover and improvements of TFP.

This table presents the estimates for the specification $\Delta TFP_{it} = \beta_0 + \beta_1 \cdot \text{Turnover}_{t-1} + \beta_2 \cdot \text{Turnover}_{t-2} + X_{it} \cdot \Gamma + \mu_j + \vartheta_t + \varepsilon_{it}$. Δ is the first-difference operator. Turnover is a dummy variable that equals one if a firm experiences a CEO turnover in the year and zero otherwise. In Model 2 we further include a Turnover dummy for the year $t-3$. The sample consists of firms in the sample of [Table 4](#), for which Execucomp data are available. Firm fixed effects and year fixed effects are controlled. Robust standard errors are clustered at the firm level. Variable definitions are in [Appendix A](#). ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively.

Variables	(1) ΔTFP	(2) ΔTFP
Turnover _{$t-1$}	0.006 [1.08]	0.006 [1.01]
Turnover _{$t-2$}	0.013** [2.39]	0.012** [1.98]
Turnover _{$t-3$}		0.004 [0.62]
Log (assets)	-0.013** [-2.57]	-0.008 [-1.61]
Tobin's q	0.039*** [13.87]	0.047*** [15.86]
Cash/assets	0.084*** [2.85]	0.081** [2.50]
Debt/assets	-0.023 [-1.40]	-0.022 [-1.27]
R&D/assets	-0.721*** [-4.46]	-0.660*** [-3.93]
Observations	22,537	19,858
R-squared	0.081	0.082
Firm FE	Y	Y
Year FE	Y	Y

- Turnover one year earlier is positive but not significant, suggesting productivity takes time to respond.
- The improvement does not reverse after three years, so the effect does not look like a short-lived accounting artifact.

5.13 Table 11: Outputs, inputs, and operating channels

Read:

- More informative stock prices raise log revenue and reduce both SG&A/assets and labor cost.
- With PIN, the coefficients are roughly 0.042 for revenue, -0.026 for SG&A, and -0.295 for labor cost; PSI gives the same signs.
- The table turns the abstract productivity result into an operating-margin result: better price discovery is associated with higher output and leaner operations.

Table 11
Outputs, inputs, and TFP improvements.

This table presents panel regressions of revenue, operating, and labor expenses on stock price informativeness and other firm-level controls. The operating cost is measured by SG&A (scaled by total assets), and the labor cost is measured by the wage expenses (xlr in Compustat). Stock price informativeness is measured by the probability of information-based trading (PIN) and stock price nonsynchronicity (PSI). In our regressions, we use the average PIN or PSI over the previous three years. The sample consists of firms in Compustat for which the stock price informativeness measures are available for the years 1994–2015. All specifications include firm and year fixed effects. Robust standard errors are clustered at the firm level. Variable definitions are in Appendix A. ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively.

Variables	(1) Log (Revenue)	(2) Log (Revenue)	(3) SG&A	(4) SG&A	(5) Log (LaborCost)	(6) Log (LaborCost)
PIN	0.042*** [2.93]		-0.026*** [-3.89]		-0.295*** [-3.77]	
PSI		0.009*** [6.24]		-0.002** [-2.34]		-0.018* [-1.91]
Log (assets)	0.410*** [56.46]	0.418*** [56.23]	-0.004* [-1.67]	-0.004** [-1.96]	0.633*** [27.85]	0.635*** [25.71]
Tobin's q	0.035*** [19.00]	0.036*** [18.79]	-0.009*** [-11.50]	-0.009*** [-11.42]	-0.006 [-0.72]	-0.007 [-0.88]
Cash/assets	-0.321*** [-18.10]	-0.329*** [-18.35]	0.027*** [3.46]	0.031*** [3.91]	-0.18 [-1.43]	-0.241* [-1.52]
Debt/assets	-0.086*** [-7.92]	-0.088*** [-7.89]	0.012*** [2.86]	0.012*** [2.81]	-0.186*** [-2.73]	-0.219*** [-2.73]
R&D/assets	0.595*** [10.81]	0.605*** [10.94]	0.465*** [13.45]	0.467*** [13.37]	3.683*** [5.03]	3.681*** [4.94]
PP&E/assets	-0.094*** [-3.77]	-0.104*** [-4.02]	0.011 [1.20]	0.015 [1.40]	0.452*** [3.52]	0.426*** [3.23]
Log (revenue _{t-1})	0.498*** [55.61]	0.494*** [54.02]	-0.017*** [-7.35]	-0.017*** [-7.00]		
Observations	63,739	60,953	63,739	60,953	7,603	7,347
R-squared	0.889	0.889	0.077	0.079	0.663	0.661
Firm FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y

Table 12
Firm characteristics, price informativeness, and productivity.

This table presents estimates of panel regressions of TFP on the interactions of firm characteristics and stock price informativeness and other firm level control variables. The dependent variable in all specifications is TFP. Stock price informativeness is measured by the probability of information-based trading (PIN) and stock price nonsynchronicity (PSI). In our regressions, we use the average PIN or PSI over the previous three years. The sample consists of firms in Compustat for which the stock price informativeness measures are available for the years 1994–2015. All controls used in Columns 3 and 4 of Table 3 are included, but for brevity their coefficients are not displayed. All specifications include firm and year fixed effects. Robust standard errors are clustered at the firm level. Variable definitions are in Appendix A. ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively.

	(1) High assets	(2) Firm age	(3) Diversified	(4) Business risk	(5) Business risk	(6) Business risk	(7) Business risk	(8) Business risk
Firm characteristic								
Firm characteristic × PIN	-0.167*** [-3.25]		-0.013*** [-2.85]		-0.144*** [-2.85]		5.950*** [4.22]	
Firm characteristic × PSI		-0.008* [-1.88]		-0.001*** [-3.38]		-0.007* [-1.88]		0.580*** [5.51]
PIN	0.335*** [8.42]		0.453*** [9.98]		0.298*** [7.66]		0.089* [1.68]	
PSI		0.022*** [6.47]		0.034*** [8.34]		0.023*** [6.57]		0.001 [0.16]
Firm characteristic	0.058*** [3.74]	0.035*** [2.44]	-0.002** [-2.15]	-0.003*** [-2.77]	-0.002 [-0.16]	-0.017 [-1.39]	-3.568*** [-8.57]	-3.663*** [-9.86]
Observations	61,363	58,700	61,045	58,402	47,428	45,620	51,377	49,212
R-squared	0.193	0.191	0.195	0.194	0.204	0.202	0.200	0.199
Other controls	Y	Y	Y	Y	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y	Y	Y

5.14 Table 12: Firm characteristics, informativeness, and productivity

Read:

- The interaction with high assets is negative, so large firms benefit less from informative prices.
- The interaction with firm age is negative, meaning older firms are less responsive.
- Diversification also weakens the effect, while business risk strengthens it.
- These patterns fit the paper's theory that flexibility and information-processing ability matter.

5.15 Table 13: Financial constraints and informativeness

Read:

- Across several financial-constraint measures, the interaction with SPI is usually positive and statistically significant.
- The broad lesson is that constrained firms appear to benefit especially strongly from better market information.

- The one weak exception is the PIN interaction with the Kaplan–Zingales index, which is close to zero.

Table 13
Financial constraints, price informativeness, and productivity.

This table presents estimates of panel regressions of TTP on the interactions of financial constraint measures and stock price informativeness and other firm-level control variables. We use four financial constraint measures: no-dividend dummy, Whited and Wu index, no bond rating dummy, and Kaplan–Zingales index. The dependent variable in all specifications is TTP. Stock price informativeness is measured by the probability of information-based trading (PIN) and stock price nonsynchronicity (PSI). In our regressions, we use the average PIN or PSI over the previous three years. The sample consists of firms in Compustat for which the stock price informativeness measures are available for the years 1994–2015. All controls used in Columns 3 and 4 of Table 2 (Panel A) are included, but for brevity their coefficients are not displayed. All specifications include firm and year fixed effects. Robust standard errors are clustered at the firm level. Variable definitions are in Appendix A. ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively.

Fin. constraint	(1) No dividend	(2)	(3) WW index	(4)	(5) Bond rating	(6)	(7) KZ index	(8)
PIN × Fin. const.	0.119*** [2.66]		0.705*** [3.05]		0.108*** [2.18]		0.005 [0.31]	
PSI × Fin. const.		0.010*** [3.02]		0.043*** [2.07]		0.008* [1.73]		0.002* [1.89]
PIN	0.151*** [4.26]		0.381*** [4.90]		0.145*** [3.86]		0.214*** [7.54]	
PSI		0.009*** [2.56]		0.023*** [3.63]		0.010*** [2.44]		0.017*** [5.62]
Fin. const.	−0.010 [−0.77]	−0.006 [−0.54]	1.122*** [12.69]	1.244*** [14.01]	0.017 [1.05]	0.025 [1.60]	−0.022*** [−5.06]	−0.026*** [−6.54]
Observations	57,394	54,834	56,774	54,229	50,783	48,497	52,524	50,119
R-squared	0.205	0.204	0.219	0.213	0.198	0.197	0.215	0.215
Other controls	Y	Y	Y	Y	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y	Y	Y

Table 14
Product market competition, stock price informativeness, and productivity.

This table presents estimates of panel regressions of TTP on the interactions of product market competition measures and stock price informativeness and other firm-level control variables. Product market competition is measured by product similarity, product market fluidity, and TNC HHI. The text-based network industry classification is used to construct these measures, which are available at the Hoberg–Phillips Data Library. In the tests, dummy variables for high competition are defined based on these competition measures: high similarity, high fluidity, and low HHI, which are based on the median of the relevant measures in a year. The dependent variable in all specifications is TTP. Stock price informativeness is measured by the probability of information-based trading (PIN) and stock price nonsynchronicity (PSI). In our regressions, we use the average PIN or PSI over the previous three years. The sample consists of firms in Compustat for which the stock price informativeness measures are available for the years 1994–2015. All controls used in Columns 3 and 4 of Table 2 (Panel A) are included, but for brevity their coefficients are not displayed. All specifications include firm and year fixed effects. Robust standard errors are clustered at the firm level. Variable definitions are in Appendix A. ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively.

Competition Measure	(1) Low HHI	(2)	(3) High similarity	(4)	(5) High fluidity	(6)
Competition × PIN	0.111*** [2.81]		0.091* [1.75]		0.044 [0.87]	
Competition × PSI		0.010*** [3.31]		0.008** [1.97]		0.006* [1.69]
PIN	0.301*** [8.58]		0.319*** [9.20]		0.336*** [9.83]	
PSI		0.023*** [7.22]		0.025*** [7.77]		0.026*** [8.27]
Competition	−0.022** [−2.26]	−0.017** [−2.41]	−0.027** [−2.14]	−0.022** [−2.19]	−0.036*** [−3.17]	−0.035*** [−4.15]
Observations	46,848	44,780	46,848	44,780	43,421	41,490
R-squared	0.363	0.360	0.362	0.360	0.371	0.368
Other controls	Y	Y	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y

5.16 Table 14: Product-market competition and informativeness

Read:

- The interaction between SPI and competitive environments is positive in most specifications.
- The evidence is strongest for low HHI and high product similarity, and somewhat weaker but still supportive for product-market fluidity.
- Competition seems to sharpen firms’ incentives to use information from prices efficiently.

5.17 Table 15: Corporate governance and the role of informativeness

Read:

- Stronger governance amplifies the SPI effect: interactions with high institutional ownership and blockholders are positive.
- The interaction with the G-index is negative for PIN, meaning weaker governance dampens the productivity payoff from informative prices.
- This reinforces the view that informative prices matter most when governance makes managers responsive to them.

Table 15
Corporate governance and the role of stock price informativeness.

This table presents estimates of panel regressions of TTP on the interactions of corporate governance measures and stock price informativeness and other firm-level control variables. The strength of corporate governance is measured by a high institutional ownership dummy (based on median in a year), the number of blockholders (logarithm), and the G-index (Gompers et al., 2003). The dependent variable in all specifications is TTP. Stock price informativeness is measured by the probability of information-based trading (PIN) and stock price nonsynchronicity (PSI). In our regressions, we use the average PIN or PSI over the previous three years. The sample consists of firms in Compustat for which the stock price informativeness measures and the governance measures are available for the years 1994–2015. All controls used in Columns 3 and 4 of Table 2 (Panel A) are included, but for brevity their coefficients are not displayed. All specifications include firm and year fixed effects. Robust standard errors are clustered at the firm level. Variable definitions are in Appendix A. ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively.

Governance measure	(1) High inst. ownership	(2)	(3) Log (N_blockholders)	(4)	(5) G-index	(6)
Governance × PIN	0.147*** [2.15]		0.138** [2.18]		−0.051*** [−3.62]	
Governance × PSI		0.020*** [3.89]		0.016*** [3.62]		−0.000 [−0.30]
PIN	0.160*** [2.73]		0.080 [1.00]		0.597*** [4.22]	
PSI		0.016** [2.56]		0.006 [0.87]		0.012 [0.78]
Governance	−0.005 [−0.37]	−0.003 [−0.21]	−0.062*** [−4.56]	−0.058*** [−4.66]	0.001 [0.17]	−0.009 [−1.49]
Observations	22,286	21,229	22,286	21,229	15,328	14,817
R-squared	0.224	0.223	0.226	0.224	0.214	0.211
Other controls	Y	Y	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y

Table 16
Alternative efficiency measures.

This table presents panel regressions of different measures of productivity/efficiency on stock price informativeness and other firm-level controls. The measures of productivity/efficiency in Columns 1 to 10 are from Loderer et al. (2016). TTP Growth (TTP Gr) in Columns 11 and 12 is from Chen et al. (2011). Stock price informativeness is measured by the probability of information-based trading (PIN) and stock price nonsynchronicity (PSI). In our regressions, we use the average PIN or PSI across the previous three years. The sample consists of firms in Compustat for which the stock price informativeness measures are available for the years 1994–2015. All specifications include firm and year fixed effects. Robust standard errors are clustered at the firm level. Variable definitions are in Appendix A. ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively.

Variables	(1) Sales/W	(2) Sales/W	(3) Sales/W	(4) Sales/W	(5) COC/Smp	(6) COC/Smp	(7) ROA	(8) ROA	(9) Loss	(10) Loss	(11) TTP Gr	(12) TTP Gr
PIN	0.042*** [3.55]	0.020** [1.90]	0.074*** [4.27]	0.170*** [2.62]	0.009*** [2.88]	0.009*** [2.88]	−0.081*** [−2.87]	−0.081*** [−2.87]	−0.136*** [−3.18]	−0.088*** [−3.18]	0.008*** [3.18]	0.008*** [3.18]
PSI	−0.550*** [−3.55]	−0.512*** [−3.55]	−0.497*** [−3.55]	−0.486*** [−3.55]	−0.486*** [−3.55]	−0.486*** [−3.55]	−0.486*** [−3.55]	−0.486*** [−3.55]	−0.486*** [−3.55]	−0.486*** [−3.55]	−0.486*** [−3.55]	−0.486*** [−3.55]
Yolton's q	0.011*** [7.62]	0.011*** [7.62]	0.007*** [7.62]	0.007*** [7.62]	0.007*** [7.62]	0.007*** [7.62]	0.007*** [7.62]	0.007*** [7.62]	0.007*** [7.62]	0.007*** [7.62]	0.007*** [7.62]	0.007*** [7.62]
Debt/assets	−0.001*** [−19.82]	−0.001*** [−19.82]	−0.001*** [−19.82]	−0.001*** [−19.82]	−0.001*** [−19.82]	−0.001*** [−19.82]	−0.001*** [−19.82]	−0.001*** [−19.82]	−0.001*** [−19.82]	−0.001*** [−19.82]	−0.001*** [−19.82]	−0.001*** [−19.82]
Cash flows	0.000*** [18.37]	0.000*** [18.37]	0.000*** [18.37]	0.000*** [18.37]	0.000*** [18.37]	0.000*** [18.37]	0.000*** [18.37]	0.000*** [18.37]	0.000*** [18.37]	0.000*** [18.37]	0.000*** [18.37]	0.000*** [18.37]
Log (sales), log	0.442*** [38.68]	0.442*** [38.68]	0.442*** [38.68]	0.442*** [38.68]	0.442*** [38.68]	0.442*** [38.68]	0.442*** [38.68]	0.442*** [38.68]	0.442*** [38.68]	0.442*** [38.68]	0.442*** [38.68]	0.442*** [38.68]
Log (employees)	−0.001*** [−11.88]	−0.001*** [−11.88]	−0.001*** [−11.88]	−0.001*** [−11.88]	−0.001*** [−11.88]	−0.001*** [−11.88]	−0.001*** [−11.88]	−0.001*** [−11.88]	−0.001*** [−11.88]	−0.001*** [−11.88]	−0.001*** [−11.88]	−0.001*** [−11.88]
Observations	63,739	60,955	63,739	60,955	63,740	60,956	63,740	60,956	41,480	41,413	53,458	50,559
R-squared	0.47	0.46	0.48	0.47	0.28	0.28	0.24	0.24	0.12	0.12	0.14	0.14
Firm FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y

5.18 Table 16: Alternative efficiency measures

Read:

- The main result survives when the dependent variable is sales/book assets, sales/value of assets in place, ROA, a loss dummy, or TFP growth.
- The only clear weak spot is the PIN coefficient for COGS per employee, which is not significant at conventional levels.
- Overall, the table shows the paper is not relying on a fragile, model-specific productivity metric.

6 Short summary

- **Conclusion.** More informative stock prices make firms more productive.
- **Mechanism.** One concrete channel runs through managerial discipline: when stock prices become less informative, CEO turnover is less sensitive to market valuation, and productivity suffers.
- **Identification lesson.** The causal evidence is strongest because three different non-fundamental shocks to informativeness—brokerage closures, S&P 500 additions, and mutual-fund redemption pressure—all point in the same direction.
- **Economic lesson.** The stock market affects the real economy not only by financing firms but also by generating information that improves resource allocation inside firms.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** Firms with more informative stock prices have higher productivity, and the evidence supports a causal interpretation.
- **Magnitude.** In the baseline regressions, a one-standard-deviation increase in PIN or PSI raises TFP by about 5% of a TFP standard deviation.
- **Causal support.** Exogenous declines in informativeness from brokerage closures, S&P 500 additions, and mutual-fund redemption pressure all reduce productivity.
- **Mechanism evidence.** Lower informativeness weakens CEO-turnover sensitivity to Tobin's q , and CEO replacement is followed by later productivity gains.
- **Operating evidence.** Informative prices are linked to higher revenues and lower overhead and labor expenses.
- **Cross-sectional implication.** The effect is stronger when firms are more flexible, more disciplined, or face stronger incentives to allocate resources well.

7.2 Economic intuition

- A stock price aggregates dispersed information from investors. When that aggregation is strong, the price tells management and the board something useful about the firm's prospects and the quality of managerial decisions.
- If the price becomes noisy because analyst coverage disappears, index comovement rises, or fund redemptions create non-fundamental trading, then the signal becomes less valuable and decision-makers rationally react less to it.
- The real consequence is not just a worse stock-market statistic; it is weaker managerial discipline and less efficient operations inside the firm.
- Informative prices matter most where firms can actually act on them: smaller, younger, focused, competitive, and well-governed firms.

7.3 Takeaway

This paper shows that stock-market informativeness has real productivity consequences. Multiple quasi-experimental shocks to informativeness move productivity in the predicted direction, implying that listed prices matter not just for financing, but also for governance and operations inside firms.